Yunhui Song

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Research Interests

I design and study interactive systems that connect the digital and physical worlds through spatiotemporal AR/VR interfaces and wearable technologies. By combining sensing and feedback, my work aims to augment human physical capabilities and foster more immersive and engaging interactions.

Education

Aarhus University, Denmark *Sept. 2024 – Oct. 2026*
M.Sc. in Computer Science
Master's Thesis: Grade 12 (Highest Danish Grade).

Chengdu University of Technology, China *Sept. 2017 – Jun. 2021*
B.Eng. in Computer Science and Technology

Academic Experience

Human-Centered Computing Section, Aarhus University *Sept. 2025 – Present*
Research Project in 4D Gaussian Splatting for Immersive VR, Supervised by ***Prof. Stefanie Zollmann***

- **Research Assistant** – Contributed to an XR rendering system based on 3D Gaussian Splatting, developing foveated rendering strategies and low-latency eye-tracking integration. Extended the system for WebXR and Unity.
- **Master project: Pano4DGS** – Developed a pipeline for 360° panoramic scene reconstruction and dynamic object composition, covering static scene modeling, 4D Gaussian-based dynamic reconstruction, and scene integration. Built a rendering and visualization framework for immersive exploration of reconstructed scenes.

Human-Centered Computing Section, Aarhus University *Nov. 2024 – Sept. 2025*
Research Intern in Intraoral User Interfaces, Supervised by ***Prof. Michael Wessely***

Peer-Reviewed Publications

* = Equal contribution = Best Paper Honorable Mention Award

Pano4DGS: From 360° Video to Dynamic 4D Gaussians for Virtual Reality.
Yunhui Song, Yanling Hua, Hannah Schieber, Kerian Varaine, Tobias Langlotz, and Stefanie Zollmann.
(Under review at IEEE VR 2027).

(Top 1.5% of submissions) TeethMark: Enabling Tongue–Tooth Interaction with Capacitive Sensing and Electrotactile Feedback using an Intraoral Wearable.
Yijing Jiang, **Yunhui Song**, Haiyang Xu, Maria Perdices Segui, Himani Deshpande, and Michael Wessely.
(Accepted to ACM UIST 2026). Acceptance rate: 20.1% (253 of 1,259 submissions)

EchoMind: Augmenting Recorded Presentations with Context-Grounded Conversational Agents and Collective Viewer Signals.
Yunhui Song*, Haiyang Xu*, Black Sun and Niklas Elmqvist.
Proceedings of the 14th Nordic Conference on Human-Computer Interaction (NordiCHI '26), 1–20.
Acceptance rate: 21.8% (110 of 504 submissions)

Demo and Poster Publications

BIOral CapCell: An Intraoral Solution for Drug Release and Monitoring via Colorimetric Sensing.
Yijing Jiang, Haiyang Xu, **Yunhui Song**, Black Sun, Junzhe Jin, and Michael Wessely.
ACM UIST 2026 Posters (To appear).

Demonstration of TeethMark: Enabling Tongue–Tooth Interaction with Capacitive Sensing and Electrotactile Feedback using an Intraoral Wearable.
Yijing Jiang, **Yunhui Song**, Haiyang Xu, Maria Perdices Segui, Himani Deshpande, and Michael Wessely.
ACM UIST 2026 Demos (To appear).

Demonstration of BIOral: Fabricating Intraoral pH Sensor for Continuous Health Monitoring.
Yijing Jiang, Junzhe Jin, **Yunhui Song**, Haiyang Xu, and Michael Wessely. 2025.
Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25).

Teaching Experience

Engineering Interactive Technologies, Aarhus University *Sept. 2025 – Jan. 2026*
Teaching Assistant

Awards and Honors

Best Paper Honorable Mention Award *2026*
TeethMark, ACM UIST 2026 **Top 1.5% of submissions**

Student Travel Grant (DKK 4,500 ≈ €600) *Sept. 2026*
Supported attendance at ACM NordiCHI 2026

Stibo Foundation Grant (DKK 14,000 ≈ €2k) *May 2025*

Outstanding Graduate of Chengdu University of Technology *Jun. 2021*

Bronze Medal, ICPC Asia Regional Contest *Oct. 2019*

Skills

- **Programming:** C++ (Modern C++17), Python, JavaScript
- **Immersive Systems:** WebXR, Unity, AR/VR Prototyping, Spatial Interaction
- **Physical Computing:** ESP32, Arduino, Wearable Sensing, Haptics, Multimodal Feedback
- **Software:** Blender, Rhino, Git, CI/CD, Linux
- **Languages:** Mandarin (Native), English (Full Professional Proficiency)